Straight Objective Type Questions

Coulomb's Law :

1. A Copper atom has 29 electrons revolving around the nucleus. A copper ball contains 4×10^{23} atoms. What fraction of the electrons be removed to give the ball a charge of $+9.6\mu C$?
   1) -1.8×10^{-13} 2) -1.3×10^{-12} 3) 6×10^{-10} 4) -5.2×10^{-12}

2. A pith ball of mass 9×10^{-5} kg carries a charge of $5\mu C$. What must be charge in another pith ball placed directly 2 cm above the given pith ball such that they are held in equilibrium ?
   1) $3.2 \times 10^{-11}C$ 2) $7.84 \times 10^{-12}C$ 3) $1.2 \times 10^{-13}C$ 4) $1.6 \times 10^{-19}C$

3. N fundamental charges each of charge 'q' are to be distributed as two point charges seperated by a fixed distance, then the maximum to minimum force bears a ratio (N is even and greater than 2)
   1) $\frac{(N-1)^2}{4N^2}$ 2) $\frac{4N^2}{(N-1)}$ 3) $\frac{N^2}{4(N-1)}$ 4) $\frac{2N^2}{(N-1)}$

4. Two point charges placed at a distance r, in the air experience a certain force, then the distance at which they will experience the same force in the medium of dielectric constant K is
   1) Kr 2) $\frac{r}{K}$ 3) $\frac{r}{\sqrt{K}}$ 4) $r\sqrt{K}$

5. Two charges 2 c and 6 c are seperated by a finite distance. If a charge of –4c is added to each of them, the initial force of 12×10^3N will change to
   1) 4×10^3N repulsion 2) 4×10^2N repulsion 3) 6×10^3N attraction 4) 4×10^3N attraction

6. A charge of $1\mu C$ is divided into two parts such that their charges are in the ratio of 2:3. These two charges are kept at a distance 1m apart in vacuum. Then, the electric force between them (in Newtons) is
   1) 0.216 2) 0.00216 3) 0.0216 4) 2.16

7. Two equal point charges each of $3\mu C$ are separated by a certain distance in metres. If they are located at $(\hat{i}+\hat{j}+\hat{k})$ and $(2\hat{i}+3\hat{j}+3\hat{k})$, then the electrostatic force between them is
   1) 9×10^3 N 2) 9×10^{-3} N 3) 10^{-3} N 4) 9×10^{-2} N

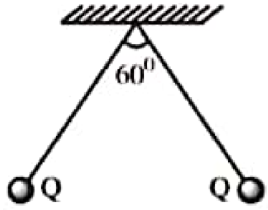
8. Under the influence of the Coulomb field of charge +Q, a charge –q is moving around it in an elliptical orbit. Find out the correct statement(s)
   1) The angular momentum of the charge –q is constant
   2) The linear momentum of the charge –q is constant
   3) The angular velocity of the charge –q is constant
   4) The linear speed of the charge –q is constant

9. A conductor has been given a charge -3×10^{-7} C by transferring electron. Mass increase (in kg) of the conductor and the number of electrons added to the conductor are respectively
   1) 2×10^{-16} and 2×10^{31} 2) 5×10^{-31} and 5×10^{19}
   3) 3×10^{-19} and 9×10^{16} 4) 2×10^{-18} and 2×10^{12}

10. Two identical charges repel each other with a force equal to 10 mg wt when they are 0.6 m apart in air. ($g = 10ms^{-2}$). The value of each charge is

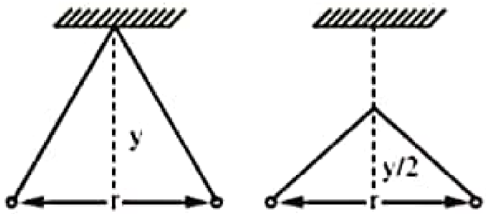
1) 2mC 2) 2×10^{-7} C 3) 2 nC 4) 2μC

11. Two small spherical balls each carrying a charge $Q = 10\mu C$ (10 micro-coulomb) are suspended by two insulating threads of equal lengths 1 m each, from a point fixed in the ceiling. It is found that in equilibrium threads are separated by an angle 60^0 between them, as shown in the figure. What is the tension in the threads (Given : $\frac{1}{(4\pi\varepsilon_0)} = 9\times10^9 Nm/C^2$)



1) 18 N 2) 1.8 N 3) 0.18 N 4) None of the above

12. Two path balls carrying equal charges are suspended from a common point by strings of equal length, the equilibrium separation between them is r. Now the strings are rigidly clamped at half the height. The equilibrium separation between the balls now become



1) $\left(\frac{2r}{3}\right)$ 2) $\left(\frac{r}{\sqrt{2}}\right)^2$ 3) $\left(\frac{r}{\sqrt[3]{2}}\right)$ 4) $\left(\frac{2r}{\sqrt{3}}\right)$

13. Two charges, each equal to q, are kept at x = –a and x = a on the x-axis. A particle of mass m and charge $q_0 = \frac{q}{2}$ is placed at the origin. If charge q_0 is given a small displacement ($y << a$) along the y-axis, the net force acting on the particle is proportional to

1) y 2) –y 3) 1/y 4) –1/y

14. Two identical conducting balls A and B have positive charges q_1 and q_2 respectively. But $q_1 \neq q_2$. The balls are brought together so that they touch each other and then kept in their original positions. The force between them is

1) less than that before the balls touched 2) greater than that before the balls touched
3) same as that before the balls touched 4) zero

15. Electric charges of 1μC, – 1μC and 2μC are placed in air at the corners A, B and C respectively of an equilateral triangle ABC having length of each side 10 cm. The resultant force on the charge at C is

1) 0.9 N 2) 1.8 N 3) 2.7 N 4) 3.6 N

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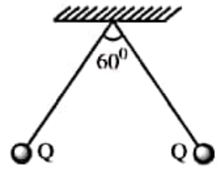

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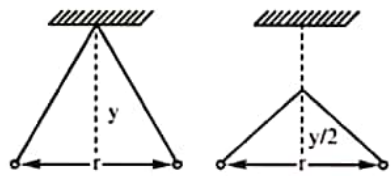

1) $\left(\frac{2r}{3}\right)$ 2) $\left(\frac{r}{\sqrt{2}}\right)^2$ 3) $\left(\frac{r}{\sqrt[3]{2}}\right)$ 4) $\left(\frac{2r}{\sqrt{3}}\right)$

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15. Electric charges of $1\mu C$, $-1\mu C$ and $2\mu C$ are placed in air at the corners A, B and C respectively of an equilateral triangle ABC having length of each side 10 cm. The resultant force on the charge at C is

1) 0.9 N 2) 1.8 N 3) 2.7 N 4) 3.6 N

Intensity of Electric Field :

16. A charge of 2C is placed on the x-axis at 1m from the origin along –ve x-axis. Infinite number of charges each of magnitude 2C are placed on x-axis at 1m, 2m, 4m, ... from origin along +ve x-axis. The first charge is positive and alternate charges are of opposite in nature. The electric field intensity at the origin

1) $\frac{1}{10\pi\epsilon_o}$ along +ve x-axis 2) $\frac{1}{10\pi\epsilon_o}$ along –ve x-axis
3) $\frac{1}{\pi\epsilon_o}$ along +ve x-axis 4) $\frac{1}{\pi\epsilon_o}$ along –ve x-axis

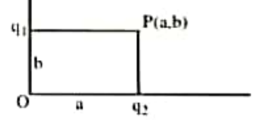
17. An electric field is acting vertically upwards. A small body of mass 1 gm and charge $-1\mu C$ is projected with a velocity 10 m/s at an angle 45° with horizontal. Its horizontal range is 2m then the intensity of electric field is :(g=10 m/s^2)

1) 20,000 N/C 2) 10,000 N/C 3) 40,000 N/C 4) 90,000 N/C

18. A bob of a simple pendulum of mass 40gm with a positive charge 4×10^{-6}C is oscillating with time period 'T_1'. An electric field of intensity 3.6×10^4 N/c is applied vertically upwards now time period is T_2. The value of 'T_2/T_1' is (g=10 m/s^2)

1) 0.16 2) 0.64 3) 1.25 4) 0.8

19. Two point changes $q_1 = 2\mu C$ and $q_2 = 1\mu C$ are placed at distances b = 1cm and a = 2cm from the origin on the y and x axes. The electric field vector at point P (a,b) will subtend an angle 'θ' with the x - axis given by



1) $\tan\theta = 1$ 2) $\tan\theta = 2$
3) $\tan\theta = 3$ 4) $\tan\theta = 4$

20. A point charge q moves from 'P' to 'S' along path PQRS in a uniform electric field E directed parallel to positive X-axis. The co-ordinates of the points P,Q, R and S are (a, b, 0), (2a, 0, 0), (0, –b, 0) and (0, 0, 0) respectively. The work done by field in the above process is given by

1) qEa 2) –qEa 3) $qEa\sqrt{2}$ 4) $qE\sqrt{(2a)^2+b^2}$

21. The surface charge density of a sphere of radius r is σ. It is placed at a point A. The electric field intensity at B due to this sphere is E. Another charged sphere of radius 2r is placed at B. If the intensity at the centre of A and B is E/2 then the surface charge density of B is

1) $\frac{\sigma}{2}$ 2) $-\frac{\sigma}{2}$ 3) $\frac{-7\sigma}{32}$ 4) $\frac{7\sigma}{32}$

22. There is a uniform electric field of strength 10^3Vm^{-1}along Y-axis. A body of mass 1gm and charge 10^{-6} C is projected into the field from origin along the positive X-axis with a velocity of 10 ms^{-1}. Its speed in ms^{-1} after 10s is (Neglect gravitation)

1) 10 2) $5\sqrt{2}$ 3) $10\sqrt{2}$ 4) 20

23. A particle of mass 1Kg and carrying 0.01 C is at rest on an inclined plane of angle 30^0 with horizontal when an electric field of $\frac{490}{\sqrt{3}}NC^{-1}$ applied parallel to horizontal, the coefficient of friction is

1) 0.5 2) $1/\sqrt{3}$ 3) $\sqrt{3}/2$ 4) $\sqrt{3}/7$

24. Four charges equal to $-Q$ are placed at the four corners of a square and a charge q is at its centre. If the system is in equilibrium, the value of q is

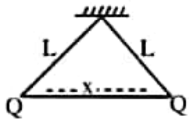
1) $-\frac{Q}{4}(1+2\sqrt{2})$ 2) $\frac{Q}{4}(1+2\sqrt{2})$ 3) $-\frac{Q}{2}(1+2\sqrt{2})$ 4) $\frac{Q}{2}(1+2\sqrt{2})$

25. If two charges +q and +4q are seperated by a distance 'd' and a point charge Q is placed on the line joining the above two charges and in between them such that all charges are in equilbrium. Then the charge Q and it's position are

1) $-\frac{4q}{9}$ at a distance $\frac{d}{3}$ from 4q 2) $\frac{-2Q}{3}$ at a distance $\frac{d}{3}$ from q

3) $-\frac{4q}{9}$ at a distance $\frac{d}{3}$ from q 4) $\frac{-2Q}{3}$ at a distance $\frac{d}{3}$ from 4q

26 Two similar balls of mass 'm' are hung by a silk thread of length L and carry similar charges Q as in figure



Assuming the separation to be small, the separation between the balls (denoted by x) is equal to

1) $\left[\frac{Q^2.2L}{4\pi\epsilon_o.mg}\right]^{\frac{1}{3}}$ 2) QLmg 3) $\frac{QL}{mg}$ 4) $\frac{Q^2L}{2\pi mg}$

27. Let $P(r)=\frac{Q}{\pi R^4}r$ be the charge density distribution for a solid sphere of radius R and total charge Q. For a point 'p' inside the sphere at distance r^1 from the centre of the sphere, the magnitude of electric field is

1) 0 2) $\frac{Q}{4\pi\epsilon_0 r_1^2}$ 3) $\frac{Qr_1^2}{4\pi\epsilon_0 R^4}$ 4) $\frac{Qr_1^2}{3\pi\epsilon_0 R^4}$

28. An α-particle fo mass 6.4×10^{-27}kg and charge 3.2×10^{-19}C is situated in a uniform electric field of $1.6\times10^5 vm^{-1}$. The velocity of the particle at the end of 2×10^{-2}m path when it starts from rest is

1) $2\sqrt{3}\times10^5 ms^{-1}$ 2) $8\times10^5 ms^{-1}$ 3) $16\times10^5 ms^{-1}$ 4) $4\sqrt{2}\times10^5 ms^{-1}$

29. An electron initially at rest falls a distance of 1.5cm in a uniform electric field of magnitude 2×10^4N/C. The time taken by the electron to fall this distance is

1) 1.3×10^2s 2) 2.1×10^{-12}s 3) 1.6×10^{-10}s 4) 2.9×10^{-9}s

30. A charged particle of mass m and charge q is releaseed from rest in a uniform electric field E. Neglecting the effect of gravity, the kinetic energy of the charged particle after 't' second is

1) $\frac{Eq^2m}{2t^2}$ 2) $\frac{2E^2t^2}{mq}$ 3) $\frac{E^2q^2t^2}{2m}$ 4) $\frac{Eqm}{t}$

Electric flux & Gauss's Law :

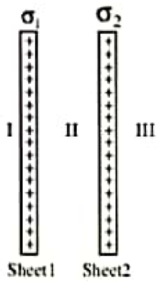
31. A cylinder of radius R and length L is placed in a uniform electric field E parallel to the cylinder axis. The total flux from the curved surface of the cylinder is given by

1) $2\pi R^2 E$ 2) $\pi R^2 / E$ 3) $\frac{(\pi R^2 - \pi R)}{E}$ 4) zero

32. A cube is arranged such that its length, breadth and height are along X, Y and Z directions. One of its corners is situated at the origin. Length of each side of the cube is 25 cm. The components of electric field are $E_x = 400\sqrt{2}$ N/C, $E_y = 0$ and $E_z = 0$ respectively. The flux coming out of the cube at one end will be

1) $25\sqrt{2}\ Nm^2/C$ 2) $5\sqrt{2}\ Nm^2/C$ 3) $250\sqrt{2}\ Nm^2/C$ 4) $25 Nm^2/C$

33. Two parallel plane sheets 1 and 2 carry uniform charge densities σ_1 and σ_2(see fig. The electric field in the region marked I is $(\sigma_1 > \sigma_2)$:



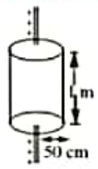
1) $-\frac{\sigma_1}{2\varepsilon_0}$

2) $-\frac{\sigma_2}{2\varepsilon_0}$

3) $\frac{(\sigma_1 - \sigma_2)}{2\varepsilon_0}$

4) $\frac{(\sigma_1 + \sigma_2)}{2\varepsilon_0}$

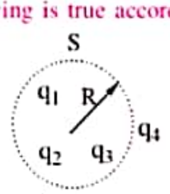
34. Electric charge is uniformly distributed along a long straight wire of radius 1 mm. The charge per cm length of the wire is Q coulomb. Another cylindrical surface of radius 50 cm and length 1 m symmetrically encloses the wire as shown in the figure. The total electric flux passing through the cylindrical surface is



1) $\frac{Q}{\varepsilon_0}$ 2) $\frac{100Q}{\varepsilon_0}$

3) $\frac{10Q}{(\pi\varepsilon_0)}$ 4) $\frac{100Q}{(\pi\varepsilon_0)}$

35. q_1, q_2, q_3 and q_4 are point charges located at points as shown in the figure and S is a spherical Gaussian surface of radius R. Which of the following is true according to the Gauss's law



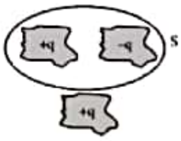
1) $\oint(\vec{E}_1 + \vec{E}_2 + \vec{E}_3).d\vec{A} = \frac{q_1 + q_2 + q_3}{\varepsilon_0}$

2) $\oint(\vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4).d\vec{A} = \frac{(q_1 + q_2 + q_3)}{\varepsilon_0}$

3) $\oint(\vec{E}_1 + \vec{E}_2 + \vec{E}_3).d\vec{A} = \frac{(q_1 + q_2 + q_3 + q_4)}{\varepsilon_0}$

4) $\oint(\vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4).d\vec{A} = \frac{(q_1 + q_2 + q_3 + q_4)}{\varepsilon_0}$

36. Shown below is a distribution of charges. The flux of electric field due to these charges through the surface S is



1) $3q/\varepsilon_0$

2) Zero

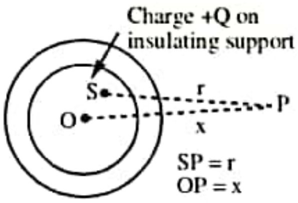
3) q/ε_0

4) $2q/\varepsilon_0$

37. An infinitely long thin straight wire has uniform linear charge density of $\frac{1}{3}cm^{-1}$. Then, the magnitude of the electric intensity at a point 18cm away is (Given $\varepsilon_0 = 8.8\times10^{-12}c^2Nm^{-2}$)

1) $0.33\times10^{11}NC^{-1}$
2) $3\times10^{11}NC^{-1}$
3) $0.66\times10^{11}NC^{-1}$
4) $1.32\times10^{11}NC^{-1}$

38. The adjacent diagram shown a charge +Q held on an insulating support S and enclosed by a hollow spherical conductor. O represents the centre of the spherical conductor and P is a point such that OP = x and SP = r. The electric field at point P will be



1) $\frac{Q}{4\pi\varepsilon_0 x^2}$
2) $\frac{Q}{4\pi\varepsilon_0 r^2}$
3) 0
4) None of the above

39. The electric charges are distributed in a small volume. The flux of the electric field through a spherical surface of radius 10 cm surrounding the total charge is 20 Vm. The flux over a concentric sphere of radius 20 cm will be

1) 20 Vm
2) 25 Vm
3) 40 Vm
4) 200 Vm

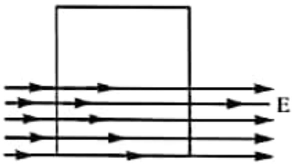
40. A sphere of radius R has a volume density of charge $\rho = kr$, where r is the distance from the centre of the sphere and k is constant. The magnitude of the electric field which exists at the surface of the sphere is given by (ε_0 = permittivity of the free space)

1) $\frac{4\pi kR^4}{3\varepsilon_0}$
2) $\frac{kR}{3\varepsilon_0}$
3) $\frac{4\pi kR}{\varepsilon_0}$
4) $\frac{kR^2}{4\varepsilon_0}$

41. The electrostatic potential inside a charged spherical ball is given by $\phi = ar^2 + b$ where r is the distance from the centre; a, b are constants. Then the charge density inside the ball is

1) $-24\pi a\varepsilon_0 r$
2) $-6a\varepsilon_0 r$
3) $-24\pi a\varepsilon_0$
4) $-6a\varepsilon_0$

42. A square surface of side L meters is in the plane of the paper. A uniform electric field $\vec{E}$ (volt/m), also in the plane of the paper, is limited only to the lower half of the square surface, (see fiture). The electric flux is SI units associated with the surface is

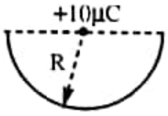


1) zero 2) EL^2 3) $EL^2/(2\varepsilon_0)$ 4) $EL^2/2$

43. In a region, the intensity of an electric field is given by $\vec{E}=2\hat{i}+3\hat{j}+\hat{k}$ in NC^{-1}. The electric flux through a surface $\vec{S}=10\hat{i}m^2$ in the region is

1) 5 Nm^2C^{-1} 2) 10 Nm^2C^{-1} 3) 15 Nm^2C^{-1} 4) 20 Nm^2C^{-1}

44. A charge 10μC is placed at the centre of hemisphere of radius R = 10 cm as shown. The electric flux through the hemisphere (in MKS units) is



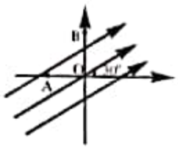
1) 20×10^5 2) 10×10^5 3) 6×10^5 4) 2×10^5

Electric Potential & Potential Energy :

45. If the electric potential of the inner metal sphere is 10 volt & that of the outer shell is 5 volt, then the potential at the centre will be :

1) 10 volt 2) 5 volt 3) 15 volt 4) 0

46. A uniform electric field of 100 V/m is directed at 30^0 with the positive x-axis as shown in figure. Find the potential difference V_B-V_A if OA = 2 m and OB = 4 m.



1) $100(2-\sqrt{3})$volt 2) $100(3-\sqrt{2})$volt 3) $-100(3+\sqrt{2})$volt 4) $-100(2+\sqrt{3})$volt

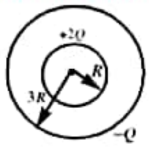
47. Find the potential function V(x, y) of an electrostatic field $\vec{E}=2axy\hat{i}+a(x^2-y^2)\hat{j}$ where 'a' is a constant.

1) $V_0+ax^2y-\frac{ay^3}{3}$ 2) $V_0-axy^2-\frac{ay^2}{3}$ 3) $V_0+axy^2+\frac{ay^3}{3}$ 4) $V_0-ax^2y+\frac{ay^3}{3}$

48. A solid metal sphere of radius R has a charge +2Q.A hollow spherical shell of radius 3R placed concentric with the first sphere has net charge -Q. Calculate the potential difference between the spheres.

1) $\frac{Q}{\pi\varepsilon_0 R}$ 2) $\frac{Q}{3\pi\varepsilon_0 R}$

3) $\frac{Q}{2\pi\varepsilon_0 R}$ 4) $\frac{Q}{7\pi\varepsilon_0 R}$



49. Two equal positive charges are kept at points A and B. The electric potential at the points between A and B (excluding these points) is studied while moving from A to B. The potential

1) contiuously increases 2) continuously decreases

3) increases then decreases 4) decreases then increases

50. Find V_{ab} in an electric field $\vec{E}=(2\hat{i}+3\hat{j}+4\hat{k})N/C$ where $\vec{r}_a=(\hat{i}-2\hat{j}+\hat{k})m$ and $\vec{r}_b=(2\hat{i}+\hat{j}-2\hat{k})m$

1) 2 volt 2) –1 volt 3) 1 volt 4) 3 volt

51. For a spherical shell

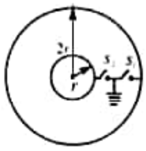
1) If potential inside it is zero then it necessarily electrically neutral

2) Electric field in a charged conducting spherical shell can be zero only when the charge is uniformly distributed.

3) Electric potential due to induced charges at a point inside it will always be zero

4) None of these

*52. There are two concentric spherical shells of radii r and 2r. Initially a charge Q is given to the inner shell. Now switch S_1 is closed and S_2 is opened then S_2 is closed and S_1 is opened and the process is repeated n times for both the keys alternatively. Find the final potential differnce between the shells.

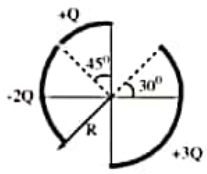


1) $\frac{1}{4^{n+1}}\left[\frac{Q}{2\pi\varepsilon_0 r}\right]$ 2) $\frac{1}{2^{n+1}}\left[\frac{Q}{4\pi\varepsilon_0 r}\right]$ 3) $\frac{1}{2^{n+1}}\left[\frac{Q}{2\pi\varepsilon_0 r}\right]$ 4) $\frac{1}{2^{n+1}}\left[\frac{Q}{\pi\varepsilon_0 r}\right]$

53. Figure shows three circular arcs, each of radius R and total charge as indicated. The net electric potential at the centre of curvature is :

1) $\frac{Q}{2\pi\varepsilon_0 R}$ 2) $\frac{Q}{4\pi\varepsilon_0 R}$

3) $\frac{2Q}{\pi\varepsilon_0 R}$ 4) $\frac{Q}{\pi\varepsilon_0 R}$



54. An infinite nonconducting sheet of charge has a surface charge density of 10^{-7} C/m^2. The separation between two equipotential surfaces near the sheet whose potential differ by 5V is

1) 0.88 cm 2) 0.88 mm 3) 0.88 m 4) 5×10^{-7} m

55. There is an electric field in +X-direction. If the work done on moving a charge of 0.2C through a distance of 2m along a line making an angle of 60° with +X-axis is 4J. The value of E is

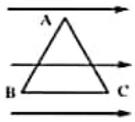
1) $\sqrt{3}NC^{-1}$ 2) $4NC^{-1}$ 3) $5NC^{-1}$ 4) $20NC^{-1}$

56. There are 27 drops of a conducting fluid. Each has a radius r and they are charged to a potential V_0. These are combined to form a bigger drop. Its potential will be:

1) V_0 2) $3V_0$ 3) $9\,V_0$ 4) $29V_0$

57. ABC is an equilateral triangle of side 2m. If $\overline{E} = 10NC^{-1}$ then $V_A - V_B$ is

1) 10V
2) –10V
3) 20V
4) – 20V



58. The charges –q, Q, –q are placed along x-axis at x=0, x=a and x=2a. If the potential energy of the system is zero, then Q:q is

1) 1 : 2 2) 2 : 1 3) 1 : 4 4) 4 : 1

59. The closest distance of approach of an α-particle travelling with a velocity V towards a stationary nucleus is 'd'. For the closest distance to become d/3 towards a stationary nucleus of double the charge the velocity of projection of the α-particle has to be

1) 6 V 2) $\sqrt{6}$ V 3) $V/_{56}$ 4) $\frac{3V}{2}$

60. Four equal charges Q are placed at the four corners of a square of each side is 'a'. Work done in moving a charge –Q from its centre to infinity

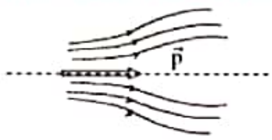
1) zero 2) $\frac{\sqrt{2}Q^2}{4\pi\epsilon_o a}$ 3) $\frac{\sqrt{2}Q^2}{\pi\epsilon_o a}$ 4) $\frac{Q^2}{2\pi\epsilon_o a}$

Electric dipole :

61. Three electric point charges q, q and - 2q are placed at the three corners of an equilateral triangle of side l. The magnitude of electric dipolemoment of the system of three particles is:

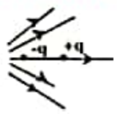
1) ql 2) $2ql$ 3) $\sqrt{3}\ ql$ 4) $4ql$

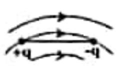
62. Consider the dipole $\vec{p}$ kept in a space of electric field a shown. The dipole will move

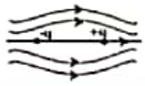


1) upwards 2) downwards 3) towards right 4) towards left

63. In which of the following cases, force acting on the dipole could be zero

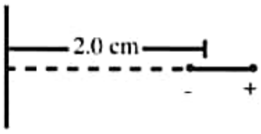
1) 

2) 

3) 

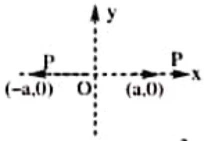
4) None

64. An electric dipole consists of charges $\pm 2.0\times10^{-8}C$ separated by a distance of $2.0\times10^{-3}m$. It is placed near a long line charge of linear charge density $4.0\times10^{-4}C/m$. as shown in figure, such that the negative charge is at a distance of 2.0 cm from the line charge. Find the force acting on the dipole



1) 0.6 N towards the line charge
2) 0.1 N away from the line charge
3) 0.9 N towards the line charge
4) 1.5 N away from the line charge

65. A system of two electric dipoles, each of dipole moment having magnitude P are arranged in the configuration shown in figure. The electrostatic interaction energy of this system of two dipoles is :

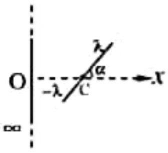


1) $\frac{5}{4}\frac{kp^2}{a^3}$
2) $\frac{7kp^2}{a^3}$
3) $\frac{kp^2}{4a^3}$
4) None of these

66. An electric point dipole is placed at the origin O with its dipolemoment along the X-axis. A point A is at a distance r from the origin such that OA makes an angle $\pi/3$ with the X-axis. If the electric field $\vec{E}$ due to the dipole at A makes an angle θ with the positive X-axis, the value of θ is

1) $\pi/3$
2) $(\pi/3)+\tan^{-1}(\sqrt{3}/2)$
3) $(\pi/3)-\tan^{-1}(\sqrt{3}/2)$
4) $\tan^{-1}(\sqrt{3}/2)$

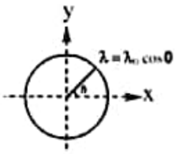
67. An infinite nonconducting sheet at x=0 carries a uniform surface charge density σ. A thin rod of length 2L has a uniform linear charge density λ on one half and $-\lambda$ on the other half. The rod is hinged at its mid point C at x=2 L and lies in the XOZ plane making an angle α with the positive X-axis as shown in figure. The torque $\vec{\tau}$ experienced by the rod is $\frac{4\sigma\lambda L^2}{k\epsilon_0}\hat{j}\sin\alpha$. The value of k is



1) 1
2) 2
3) 3
4) 8

68. A ring of radius R carries a non-uniform charge of linear density $\lambda=\lambda_0\cos\theta$ (see in the figure) Magnitude of the net dipolement of the ring is:

1) $\pi R^2\lambda_0$
2) $2\pi R^2\lambda_0$
3) $\frac{\pi R^2}{2}\lambda_0$
4) $4\pi R^2\lambda_0$

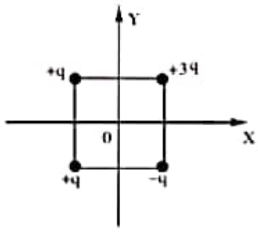


*69. An electric field line emerges from a positive point charge +q at angle α to the straight line connecting it to a negative point charge $-2q$ as shown in figure. At what angle β will the field line enter the charge $-2q$?

1) α 2) $2\text{Sin}^{-1}\left(\frac{1}{\sqrt{2}}\sin\frac{\alpha}{2}\right)$

3) $\sin^{-1}\left(\frac{1}{\sqrt{2}}\sin\frac{\alpha}{2}\right)$ 4) $\sin^{-1}\left(\frac{1}{2\sqrt{2}}\sin\frac{\alpha}{2}\right)$

*70. Four point charges are held fixed at the corners of a square of side L as shown in figure. The square has its centre at the origin and sides parallel to the coordinate axes in the XOY plane. The moment of charge configuration is a null vector about the point



1) (L/2, 0)

2) (-L/2, 0)

3) (0, -L/2)

4) (0, L/2)

71. A given charge is situated at a certain distance from an electric dipole in the end-on position experiences a force F. If the distance of the charge is doubled, the force acting on the charge will be

1) 2F 2) F/2 3) F/4 4) F/8

72. An electric dipole consisting of two opoosite charges of 2×10^{-6} C each separated by a distance of 3 cm is placed in an electric field of 2×10^{5} N/C. The maximum torque on the dipole will be

1) 12×10^{-1} Nm 2) 12×10^{-3} Nm 3) 24×10^{-1} Nm 4) 24×10^{-3} Nm.

73. The electric intensity due to a dipole of length 10 cm and having a charge of $500\mu C$, at a point on the axis at a distance 20 cm from one of the charges in air, is

1) 6.25×10^{7} N/C 2) 9.28×10^{7} N/C 3) 1.31×11^{11} N/C 4) 20.5×10^{7} N/C

74. The distance between H^{+} and Cl^{-} ions in HCl molecule is $1.28\mathring{A}$. What will be the potential due to this dipole at a distance of $12\mathring{A}$ on the axis of dipole

1) 0.13 V 2) 1.3 V 3) 13 V 4) 130 V

Numerical Value type Questions

75. How much positive and negative charge is there in a cup (250 ml) of water _____ 10^{7} coulombs

76. Two charges $+20\mu c$ and $-20\mu c$ are held 1 cm apart. Calculate electric field ata point on the equatorial lie ata distance of 50 cm from the dipole in kilo newton percoulomb

77. An infinite line of charge produces a field of 9×10^{9} N/C at a distance of 0.02m. Calculate the linear charge density.

78. A wire is bent in a circle of radius 10cm. It is given a charge of $250\mu C$ which spreads uniformly the electric potential at the centre in 10^{7} volt.

79. An electric dipole when held at $30°$ with respect to a uniform electric field of 10^{4} N/C experience a torque of 9×10^{-26} N–m. Dipole moment of the dipole in 10^{-29} C-m.

EXERCISE-II

Coulomb's Law :

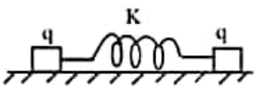
1. Two electrons separated by distance 'r' experience a force F between them. The force between a
   1) 4F 2) 2F 3) F/2 4) F/4

2. Two charges of equal magnitudes and at a distance 'r' exert a force F on each other. If the charges are halved and distance between them is doubled, then the new force acting on each charge is
   1) F/8 2) F/4 3) 4F 4) F/16

3. Two identical copper spheres are separated by 1m in vacuum. How many electrons would have to be removed from one sphere and added to the other so that they now attract each other with a force of 0.9 N ?
   1) 6.25×10^{15} 2) 62.5×10^{15} 3) 6.25×10^{13} 4) 0.65×10^{13}

4. Two positive charges seperated by a distance 2m each other with a force of 0.36 N. If the combined charge is $26\mu C$, the charges are
   1) $20\mu C, 6\mu C$ 2) $16\mu C, 10\mu C$ 3) $18\mu C, 8\mu C$ 4) $13\mu C, 13\mu C$

5. Two identical particles of charge q each are connected by a mass less spring of force constant K. They are placed over a smooth horizontal surface. They are released when the separation between them is r and spring is in it's natural length. If maximum extension of the spring is r, the value of K is (neglect gravitational effect)



   1) $\frac{q}{4r}\sqrt{\frac{1}{\pi\varepsilon_0 r}}$ 2) $\frac{q}{2r}\sqrt{\frac{1}{\pi\varepsilon_0 r}}$
   3) $\frac{2q}{r}\sqrt{\frac{1}{\pi\varepsilon_0 r}}$ 4) $\frac{q}{r}\sqrt{\frac{1}{\pi\varepsilon_0 r}}$

6. The charges on two spheres are $+7\mu C$ and $-5\mu C$ respectively. They experience a force F. If each of them is given and additional charge of $-2\mu C$, the new force of attraction will be
   1) F 2) F/2 3) $F/\sqrt{3}$ 4) 2F

7. The ratio of electrostatic and gravitational forces acting between electron and proton separated by a distance 5×10^{-11} m, will be (Charge on electron = 1.6×10^{-19} C, mass of electron = 9.1×10^{-31} kg, mass of electron = 9.1×10^{-31} kg, mass of proton = 1.6×10^{-27} kg, G = 6.7×10^{-11} Nm^2/kg^2)
   1) 2.36×10^{39} 2) 2.36×10^{40} 3) 2.34×10^{41} 4) 2.34×10^{42}

8. Two charges $+6\mu C$ and $+15\mu C$ are placed along the x-axis at x = 0 and x = 2m respectively. A negative charge is placed between them such that the resultant force on it is zero. The negative charge is placed at
   1) x = 0.775 m 2) x = 1.2 m
   3) x = 0.5 m 4) position depends on the amount of charge

9. Equal charges q are placed at the four corners A, B, C, D of a square of length a. The magnetude of the force on the charge at B will be
   1) $\frac{3q^2}{4\pi\varepsilon_0 a^2}$ 2) $\frac{4q^2}{4\pi\varepsilon_0 a^2}$ 3) $\left(\frac{1+2\sqrt{2}}{2}\right)\frac{q^2}{4\pi\varepsilon_0 a^2}$ 4) $\left(2+\frac{1}{\sqrt{2}}\right)\frac{q^2}{4\pi\varepsilon_0 a^2}$

10. When a glass rod is rubbed with silk, it

1) Gains electrons from silk
2) Gives electrons to silk
3) Gains protons from silk
4) Gives protons to silk

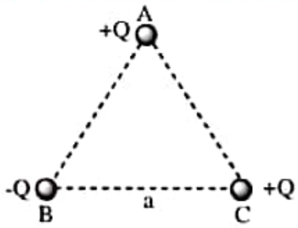
11. A body gas –80 micro coulomb of charge. Number of additonal electrons in it will be

1) 8×10^{-5}
2) 80×10^{-17}
3) 5×10^{14}
4) 1.28×10^{-17}

12. Two particles of equal mass m and charge q are placed at a distance of 16 cm. They do not experience any force. The value of $\frac{q}{m}$ is

1) l
2) $\sqrt{\frac{\pi\varepsilon_0}{G}}$
3) $\sqrt{\frac{G}{4\pi\varepsilon_0}}$
4) $\sqrt{4\pi\varepsilon_0 G}$

13. Three charges are placed at the vertices of an equilateral triangle of side 'a' as showin in the following figure. The force experienced by the charge placed at the vertex A in a direction normal to BC is



1) $Q^2/(4\pi\varepsilon_0 a^2)$
2) $-Q^2/(4\pi\varepsilon_0 a^2)$
3) zero
4) $Q^2/(2\pi\varepsilon_0 a^2)$

14. Two charges placed in air repel each other by a force of 10^{-4}N. When oil is introduced between the charges, the force becomes 2.5×10^{-5} N. The dielectric constant of oil is

1) 2.5
2) 0.25
3) 2.0
4) 4.0

15. Two copper balls, each weighting 10 g are kept in air 10 cm apart. If one electron from every 10^6 atoms is transferred from one ball to the other, the coulomb force between them is (atomic weight of copper is 63.5)

1) 2.0×10^{10} N
2) 2.0×10^{4} N
3) 2.0×10^{8} N
4) 2.0×10^{6} N

Intensity of Electric Field :

16. Two point charges Q and –3Q are placed at some distance apart. If the electric field at the location of Q is $\vec{E}$, the field at the location of –3Q is

1) $\vec{E}$
2) $-\vec{E}$
3) $+\frac{\vec{E}}{3}$
4) $-\frac{\vec{E}}{3}$

17. The electric field at (30, 30) cm due to a charge of –8 nC at the origin in NC^{-1} is

1) $-400\,(\hat{i}+\hat{j})$
2) $400\,(\hat{i}+\hat{j})$
3) $-200\sqrt{2}\,(\hat{i}+\hat{j})$
4) $200\sqrt{2}\,(\hat{i}+\hat{j})$

18. Two charges 4×10^{-9} C and -16×10^{-9} C are separated by a distance 20 cm in air. The position of the neutral point from the small charge is

1) 40/3cm
2) 20/3cm
3) 20 cm
4) 10/3cm

19. The number of electrons to be put on a spherical conductor of radius 0.1 m to produce an electric field of 0.036 N/C just above its surface is

1) 2.7×10^5
2) 2.6×10^5
3) 2.5×10^5
4) 2.4×10^5

20. The magnitude of electric intensity at a distance 'x' from a charge 'q' is E. An identical charge is placed at a distance '2x' from it. Then the magnitude of the force it experience is

1) Eq 2) 2Eq 3) $\frac{Eq}{2}$ 4) $\frac{Eq}{4}$

21. A sphere of mass 50gm is suspended by a string in an electric field of intensity $5NC^{-1}$ acting vertically upward. If the tension in the string is 520 millinewton, the charge on the sphere is ($g = 10ms^{-2}$)

1) -4×10^{-3} C 2) 4×10^{-3} C 3) 8×10^{-3} C 4) -8×10^{-3} C

22. A and B are two points separated by a distance 5cm. Two charges 10μC and 20μC are placed at A and B. The resultant electric intensity at a point P outside the charges at a distance 5cm from 10μC is

1) 54×10^6 N/Caway from 10μC 2) 56×10^6 N/C towards 10μC

3) 9×10^6 N/C away from 10μC 4) zero

23. At the corners A, B, C of a square ABCD, charges 10mC, –20mC and 10mC are placed. The electric intensity at the centre of the square to become zero, the charge to be placed at the corner D is

1) –20mC 2) +20mC 3) 30mC 4) –30mC

24. A charged oil drop is suspended in uniform field of 3×10^4 V/m so that it neither falls nor rises. The charge on the drop will be (take the mass of the charge = 9.9×10^{-15} kg & $g=10\ m/s^2$)

1) 3.3×10^{-18} C 2) 3.2×10^{-18} C 3) 1.6×10^{-18} C 4) 4.8×10^{-18} C

25. A particle of mass 'm' and charge q is placed at rest in a uniform electric field E and then released. The K.E. attained by the particle after moving a distance y is

1) qEy^2 2) qE^2y 3) qEy 4) q^2Ey

26. A proton and an $\alpha -$particle start from rest in a uniform electric field, then the ratio of times of flight to travel same distance in the field is

1) $\sqrt{5}:\sqrt{2}$ 2) $\sqrt{3}:1$ 3) 2 : 1 4) $1:\sqrt{2}$

27. In a regular hexgon each corner is at a distance 'r' from the centre. Identical charges of magnitude 'Q' are placed at 5 corners. The field at the centre is $\left(K = \frac{1}{4\pi\epsilon_0}\right)$

1) KQ/r^2 2) $\frac{6KQ}{r^2}$ 3) $\frac{5KQ}{r^2}$ 4) Zero

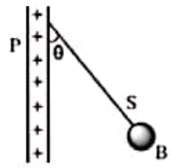
28. A point charge 50μC is located in the XY plane at the point of position vector $\vec{r}_0 = 2\hat{i}+3\hat{j}$. What is the electric field at the point of position vector $\vec{r} = 8\hat{i}-5\hat{j}$

1) 1200V/m 2) 0.04V/m 3) 900V/m 4) 4500 V/m

29. Four identical charges Q are fixed at the four corners of a square of side a. Find the electric field at a point P located symmetrically at a distance $\frac{a}{\sqrt{2}}$ from the centre of the square.

1) $\frac{Q}{2\sqrt{2}\pi\varepsilon_0 a^2}$ 2) $\frac{Q}{\sqrt{2}\pi\varepsilon_0 a^2}$ 3) $\frac{2\sqrt{2}Q}{\pi\varepsilon_0 a^2}$ 4) $\frac{\sqrt{2}Q}{\pi\varepsilon_0 a^2}$

30. A charged ball B hangs from a silk thread S, which makes an angle θ with a large charged conducting sheet P, as shown in the figure. The surface charge density σ of the sheet is proportional to



1) $\sin\theta$ 2) $\tan\theta$ 3) $\cos\theta$ 4) $\cot\theta$

Electric flux & Gauss's Law :

31. The electric field in a region of space is given by $E = 5i + 2j\ N/C$. The electric flux due to this field through an area $2m^2$ lying in the YZ plane, in S.I. units, is
1) 10 2) 20 3) $10\sqrt{2}$ 4) $2\sqrt{29}$

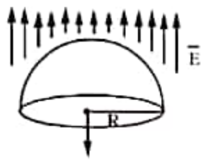
32. A charge Q is situated at the centre of a cube. The electric flux through one of the faces of the cube is
1) Q/ε_0 2) $Q/2\varepsilon_0$ 3) $Q/4\varepsilon_0$ 4) $Q/6\varepsilon_0$

33. Electric field due to an infinite sheet of charge having surface density σ is E. Electric field due to an infinite conducting sheet of same surface density of charge is
1) E/2 2) E 3) 2E 4) 4E

34. The magnitude of the electric field on the surface of a sphere of radius r having a uniform surface charge density σ is
1) σ/ε_0 2) $\sigma/2\varepsilon_0$ 3) $\sigma/\varepsilon_0 r$ 4) $\sigma/2\varepsilon_0 r$

35. If the electric flux entering and leaving an enclosed surface respectively is ϕ_1 and ϕ_2, the electric charge inside the surface will be
1) $(\phi_2 - \phi_1)\varepsilon_0$ 2) $(\phi_1 + \phi_2)/\varepsilon_0$ 3) $(\phi_2 - \phi_1)/\varepsilon_0$ 4) $(\phi_1 + \phi_2)\varepsilon_0$

36. If a hemispherical body is placed in a uniform electric field E then the flux linked with the curved surface is
1) $2\pi R^2 E$
2) $\pi R^2 E$
3) $4\pi R^2 E$
4) $6\pi R^2 E$



37. The electric flux through a Gaussian surface that encloses three charges given by $q_1 = -14$ nC, $q_2 = 78.85$ nC, $q_3 = -56$ nC
1) $10^3\ Nm^2\ C^{-1}$ 2) $10^3\ CN^{-1}\ m^{-2}$ 3) $6.32 \times 10^3\ Nm^2\ C^{-1}$ 4) $6.32 \times 10^3\ CN^{-1}\ m^{-2}$

38. An infinitely long thin straight wire has uniform linear charge density of 1/3 coul.m^{-1}. Then the magnitude of the electric intensity at a point 18 cm away is
1) $0.33 \times 10^{11}\ NC^{-1}$ 2) $3 \times 10^{11}\ NC^{-1}$ 3) $0.66 \times 10^{11}\ NC^{-1}$ 4) $1.32 \times 10^{11}\ NC^{-1}$

39. Let $\rho(r)=\frac{Q}{\pi R^4}r$ be the charge density distribution for a solid sphere of radius R and total charge Q. For a point 'p' inside the sphere at distance r_1 from the centre of the sphere, the magnitude of electric field is

1) $\frac{Q}{4\pi\epsilon_0 r_1^2}$ 2) $\frac{Qr_1^2}{4\pi\epsilon_0 R^4}$ 3) $\frac{Qr_1^2}{3\pi\epsilon_0 R^4}$ 4) 0

40. Total electric flux coming out of a unit positive charge put in air is

1) ε_0 2) ε_0^{-1} 3) $(4P\varepsilon_0)^{-1}$ 4) $4\pi\varepsilon_0$

41. The electric field in a certain region is acting radially outward and is given by E = Ar. A charge contained in a sphere of radius 'a' centred at the origin of the field, will given by

1) $A\epsilon_0 a^2$ 2) $4\pi\epsilon_0 Aa^3$ 3) $\epsilon_0 Aa^3$ 4) $4\pi\epsilon_0 Aa^3$

42. The electric intensity due to an infinite cylinder of radius R and having charge q per unit length at a distance $r(r > R)$ from its axis is

1) directly proportional to r^2
2) directly proportional to r^3
3) inversely proportional to r
4) inversely proportional to r^2

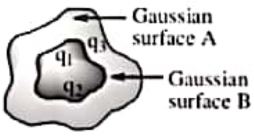
43. A sphere of radius R has a uniform distribution of electric charge in its volume. At a distance x from its centre, for x < R, the electric filed is directly proportional to

1) $1/x^2$ 2) $1/x$ 3) x 4) x^2

44. A charge Q is enclosed by a Gaussian spherical surface of radius R. If the radius is doubled, then the outward electric flux will

1) be doubled
2) increases four times
3) be reduced to half
4) remain the same

45. The electric flux for Gaussian surface A that enlcose the charged particles in free space is (given $q_1 = -14$ nC, $q_2 = 78.85$ nC, $q_3 = -56$ nC)



1) 10^3 Nm^2 C^{-1} 2) 10^3 CN^{-1} m^{-2} 3) 6.32×10^3 Nm^2 C^{-1} 4) 6.32×10^3 CN^{-1} m^{-2}

Electric Potential & Potential Energy :

46. Let V be electric potential and E the magnitude of the electric field. At a given position, which of the statement is true?

1) E is always zero where V is zero
2) V is always zero where E is zero
3) E can be zero where V is non zero
4) E is always nonzero where V is nonzero

47. Two electric charges of $9\mu C$ and $-3\mu C$ are placed 0.16 m apart in air. There will be a point P at which electric potential is zero on the line joining the two charges and in between them. The distance of P from $9\mu C$ charge is

1) 0.14 m 2) 0.12 m 3) 0.08 m 4) 0.06 m

48. Charges $5\mu C$, $-2\mu C$, $3\mu C$ and $-9\mu C$ are placed at the corners A, B,C and D of a square ABCD of side 1m. The net electric potential at the centre of the square is

1) -27 KV 2) $-27\sqrt{2}$ KV 3) -90KV 4) zero

49. An electric charge $10^{-3}\mu C$ is placed at the origin (0, 0) of X-Y coordinate system. Two points A and B are situated at $(\sqrt{2}, \sqrt{2})$ and (2, 0) respectively. The potential difference between the points A and B will be

1) 9 V 2) zero 3) 2 V 4) 4.5 V

50. In experiment, an oil drop carrying a charge Q is held stationary by a potential difference of 2400 V between the plates. To keep a drop of half the radius stationary the potential difference had to be made 600 volt. What is the charge on the second drop

1) $\frac{Q}{4}$ 2) $\frac{Q}{2}$ 3) Q 4) $\frac{3Q}{2}$

51. Two parallel plates separated by a distance of 5mm are kept at a potential diffeence of 50V. A particle of mass 10^{-15} kg and charge 10^{-11}C enters in it with a velocity 10^7m/s. The acceleration of the particle will be

1) 10^8 m/s^2 2) 5×10^5 m/s^2 3) 10^5 m/s^2 4) 2×10^3 m/s^2

52. A cloud is at a potential of 8×10^6 volt relative to the ground. A charge of 80 coulomb is transferred in lightening stroke between the cloud and the ground. Assuming the potential of the cloud to remain constant, the energy dissipated is

1) 6.4×10^8 joule 2) 6.4×10^5 joule 3) 10^5 joule 4) 10^7 joule

53. Surface charge density of a sphere of a radius 10 cm is 8.85×10^{-8}c/m^2. Potential at the centre of the sphere is

1) 1000 V 2) 885 V 3) 10^{-3} V 4) 442.5 V

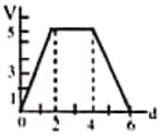
54. Two conducting spheres of radii R_1 and R_2 are at the same potential. The electric intensities on their surfaces are in the ratio of

1) 1 : 1 2) $R_1 : R_2$ 3) $R_2 : R_1$ 4) $R_1^2 : R_2^2$

55. The electric potential (V) in volts varies with x (in metre) according to the relation $V=5+4x^2$. The force experienced by a negative charge of $2\mu C$ located at x = 0.5m is

1) 2×10^{-6}N 2) 4×10^{-6}N 3) 6×10^{-6}N 4) 8×10^{-6}N

56. The variation of electric potential with distance d from a fixed point is as shown in the figure. The electric intensity at d = 5m is



1) 2.5 Vm^{-1} 2) -2.5 Vm^{-1} 3) 0.4 Vm^{-1} 4) -0.4V m^{-1}

57. The electric potential decreases uniformly from 120 V to 80 V as one moves on the X-axis from $x = -1$ cm to $x = +1$ cm. The electric field at the origin

1) must be equal to 20 V/cm
2) may be equal to 20 V/cm
3) may be greater than 20 V/cm
4) may be less than 20 V/cm

58. A charge of 5C experiences a force of 5000N when it is moved in a uniform electric field. The potential difference between two points separated by a distance of 1cm is

1) 10V 2) 250V 3) 1000V 4) 2500V

59. Electric potential inside hollow charged sphere of radius 'r' is ________ $[k=\frac{1}{4\pi\varepsilon_0}]$

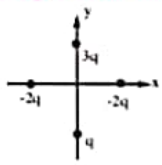
1) zero 2) $k\frac{q}{r}$ 3) $k\frac{q}{r^2}$ 4) $k\frac{q^2}{r}$

60. Two points P and Q are maintained at the potentials of 10V and –4V, respectively. The work done in moving 100 electrons from P to Q is

1) 9.60×10^{-17} J 2) -2.24×10^{-16} J 3) 2.24×10^{-16} J 4) -9.60×10^{-17} J

Eectric dipole

61. 4 charges are placed each at a distance 'a' from origin. The distance 'a' from origin. The dipole moment of configuration is :

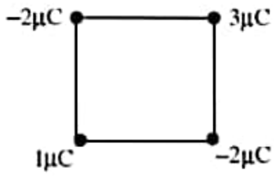


1) $2qa\hat{j}$
2) $3qa\hat{j}$
3) $2qa(\hat{i}+\hat{j})$ 4) $qa\hat{j}$

62. A dipole of dipole moment $\vec{p}=P_0\hat{j}$ is placed at point $(l,0)$. There exists an electric field $\vec{E}=2ax^2\hat{i}+(2by^2+2cy)\hat{j}$. Then

1) force on dipole is $2p_0c\hat{j}$
2) force on dipole is $p(2c+4bl)\hat{j}$
3) Torque on dipole about its centroidal axis is $2p_0a\hat{k}$
4) Torque on dipole about its centroidal axis is zero

63. Four point charges $1\mu C, -2\mu C, 3\mu C$ and $-2\mu C$ are arranged on the four vertices of a square of side 1 cm. The dipole moment of this charge assembly is



1) zero
2) $\sqrt{2}\times10^{-8}$C.m
3) $2\sqrt{2}\times10^{-8}$C.m
4) 2×10^{-8}C.m

64. Two point dipoles $p\hat{k}$ & $\frac{p}{2}\hat{k}$ are located at (0, 0, 0) & (1 m, 0, 2 m) respectively. The resultant electric field due to the two dipoles at the point (1 m, 0, 0) is

1) $\frac{9p}{32\pi\epsilon_0}\hat{k}$ 2) $\frac{-7p}{32\pi\epsilon_0}\hat{k}$ 3) $\frac{7p}{32\pi\epsilon_0}\hat{k}$ 4) none of these

65. Two electric dipoles of moment P and 64 P are placed in opposite direction on a line at a distance of 25 cm. The electric field will be zero at point between the dipoles whose distance from the dipole of moment P is

1) 5 cm 2) $\frac{25}{9}$ cm 3) 10 cm 4) $\frac{4}{13}$ cm

66. Two charges $+3.2 \times 10^{-19}$ C and -3.2×10^{-9}C kept 2.4. A° apart forms a dipole. If it is kept in uniform electric field of intensity 4×10^5 volt/m then what will be its electrical energy in equilibrium

1) $+3 \times 10^{-23}$ J 2) -3×10^{-23} J 3) -6×10^{-23} J 4) -2×10^{-23} J

67. An electric dipole of length 1 cm is placed with the axis making an angle of 30 to an electric field of strength 10^4 NC^{-1}. If it experiences a torque of $10\sqrt{2}$ Nm, the potential energy of the dipole is

1) 0.245 J 2) 2.45 J 3) 0.0245 J 4) 24.5 J

68. A sample of HCl gas is placed in an electric field of 3×10^4 NC^{-1}. The dipole moment of each HCl molecule is 6×10^{-30} c × m. The maximum torque that can act on a molecule is

1) 2×10^{-34} Nm 2) 2×10^{-34} Nm

3) 18×10^{-26} Nm 4) 0.5×10^{34} $N^{-1}m^{-1}$

69. An electric dipole is placed at an angle of 30° with an electric field intensity 2×10^5 N/C. It experiences a torque equal to 4 N m. The charge on the dipole, if the dipole length is 2 cm, is

1) 7μC 2) 8 mC 3) 2 mC 4) 5 mC

70. An electric dipole has a fixed dipole moment $\vec{p}$, which makes angle θ with respect to x-axis. When subjected to an electric field $\vec{E}_1 = E\hat{i}$, it experiences a torque $\vec{T}_1 = \tau\hat{k}$. When subjected to another electric field $\vec{E}_2 = \sqrt{3}E_1\hat{j}$ it experiences a torque $\vec{T}_2 = -\vec{T}_1$. The angle θ is

1) 90° 2) 30° 3) 45° 4) 60°

71. A point Q lies on the perpendicular bisector of an electical dipole of dipole moment p. If the distance of Q from the dipole is r (much larger than the size of the dipole), then electric field at Q is proportional to

1) p^{-1} and r^{-2} 2) p and r^{-2} 3) p^2 and r^{-3} 4) p and r^{-3}

72. The electric field due to an electric dipole at a distance r from its centre in axial position is E. If the dipole is rotated through an angle of 90° about its perpendicular axis, the electric field at teh same point will be

1) E 2) E/4 3) E/2 4) 2E

73. The electric dipole moment of an electron and a proton 4.3 nm apart is

1) 6.8×10^{-28}c-m 2) 2.56×10^{-29}c-m 3) 3.72×10^{-14}c-m 4) 11×10^{-46}c-m

74. A water molecule has an electric dipole moment 6.4×10^{-30}cm when it is in vapour state. The distance in metre between the centre of positive and negative charge of the molecule is

1) 4×10^{-10} 2) 4×10^{-11} 3) 4×10^{-12} 4) 4×10^{-13}

75. A molecule with a dipole moment p is placed in an electric field of strength E. Initially the dipole is aligned parallel to the field. If the dipole is to be rotated to be anti-parallel to the field, the work required to be done by an external agency is

1) –2pE 2) –pE 3) pE 4) 2pE

Numerical Value Type Questions

76. Calculate the Coluomb force between two alpha particles separated by a distance of 3.2×10^{-15} m in air.

77. The electric potential in a region is represented as $V = 2x + 3y - Z$. The magnitude of electric field strength is

78. A rectangular surface of sides 10 cm and 15 cm is placed inside a uniform electric field of 25 Vm^{-1}, such that normal to t he surface makes an angle of $60°$ with the direction of electric field. Find the flux of electric field through the rectangular surface.

79. Charges +q, –4q and +2q are arranged at the corners of an equilateral triangle of side 0.15m. If $q = 1\mu c$, their mutual potential energy is

80. The electric potential in volts due to an electic dipole of dipole moment 1×10^{-8} C–m at a distance of 3 m on a line making an angle of $30°$ with the axis of dipole is

KEY SHEET

EXERCISE-I

1) 4	2) 2	3) 3	4) 3	5) 4	6) 2	7) 2	8) 1	9) 4	10) 4
11) 2	12) 3	13) 1	14) 2	15) 2	16) 1	17) 3	18) 3	19) 2	20) 2
21) 4	22) 3	23) 4	24) 2	25) 3	26) 1	27) 3	28) 4	29) 4	30) 3
31) 4	32) 1	33) 4	34) 2	35) 2	36) 2	37) 1	38) 1	39) 1	40) 4
41) 4	42) 1	43) 4	44) 3	45) 1	46) 4	47) 4	48) 2	49) 4	50) 2
51) 4	52) 2	53) 1	54) 2	55) 4	56) 3	57) 2	58) 3	59) 2	60) 3
61) 3	62) 4	63) 3	64) 1	65) 3	66) 2	67) 4	68) 1	69) 1	70) 4
71) 4	72) 2	73) 1	74) 1	75) 1.338	76) 14.4	77) 0.01	78) 2.25	79) 1.8	

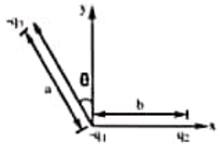
EXERCISE-II

1) 4	2) 4	3) 3	4) 2	5) 2	6) 1	7) 1	8) 1	9) 3	10) 2
11) 3	12) 4	13) 3	14) 4	15) 3	16) 3	17) 3	18) 3	19) 3	20) 4
21) 1	22) 1	23) 1	24) 1	25) 3	26) 4	27) 1	28) 4	29) 2	30) 2
31) 1	32) 4	33) 3	34) 1	35) 1	36) 2	37) 1	38) 1	39) 2	40) 2
41) 2	42) 3	43) 3	44) 4	45) 1	46) 3	47) 2	48) 2	49) 2	50) 2
51) 1	52) 1	53) 1	54) 3	55) 4	56) 1	57) 1	58) 1	59) 2	60) 3
61) 1	62) 1	63) 2	64) 2	65) 1	66) 2	67) 4	68) 3	69) 3	70) 4
71) 4	72) 3	73) 1	74) 2	75) 4	76) 90 N	77) 3.74	78) 0.1875 Nm^2C^{-1}		
79) 0.6 J	80) 8.66 V								

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. Three charges $-q_1$, $+q_2$ and $-q_3$ are placed as shown in the figure. The x-component of the force on $-q_1$ is proportional to



a) $\frac{q_2}{b^2}-\frac{q_3}{a^2}\cos\theta$ b) $\frac{q_2}{b^2}+\frac{q_3}{a^2}\sin\theta$

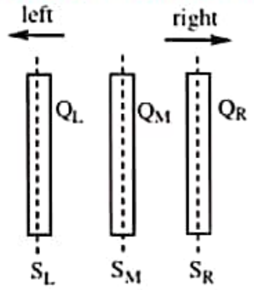
c) $\frac{q_2}{b^2}+\frac{q_3}{a^2}\cos\theta$

d) $\frac{q_2}{b^2}-\frac{q_3}{a^2}\sin\theta$

2. ABC is a right angled triangle in which AB=3cm and BC=4cm and right angle is at B .The three charge,$+15\mu C$,$+12\mu C$ and $-20\mu C$ are placed respectively at A,B andC. The force acting on B is

a) 1250 N b) 3500 N c) 1200 N d) 2250N

3. Three large identical conducting plates of area A are closely placed parallel to each other as shown (the area A is perpendicular to plane of diagram). The net charge on left , middle and right plates are Q_L, Q_M and Q_R respectively. Three infinitely large parallel surfaces S_L,S_M and S_R are drawn passing through middle of each plate such that surfaces are perpendicular to plane of diagram as shown. Then pick up the correct option(s).



a) The net charge on left side of surface S_L is equal to net charge on right side of surface S_R.

b) The net charge on left side of surface S_L is equal to net charge on right side of surface S_M

c) The net charge on left side of surface S_L is equal to net charge on right side of surface S_L

d) The net charge on right side of surface S_L is equal to net charge on left side of surface S_R.

4. A charge is placed at centre of circular face of cylinder of radius 'r' and length 'r' flux through remaining curved surface is (other than opposite circular face)

a) $\frac{q}{2\varepsilon_0}-\frac{q}{2\varepsilon_0}\left[1-\frac{1}{\sqrt{2}}\right]$ b) $\frac{q}{2\varepsilon_0}\left[1-\frac{1}{\sqrt{2}}\right]$ c) $\frac{q}{2\varepsilon_0}-\frac{q}{\varepsilon_0}\left[\frac{1}{\sqrt{2}}\right]$ d) $\frac{q}{\varepsilon_0}\left[1-\frac{1}{\sqrt{2}}\right]$

5. Electric flux through a surface of area 100 m² lying in the x, y plane is (in V-m) if $\vec{E}=\hat{i}+\sqrt{2}\hat{j}+\sqrt{3}\hat{k}$

a) 100 b) 141.4 c) 173.2 d) 200

6. A cylinder of length 2a and radius 'a' has the x-axis as its axis. Its two ends (plane surfaces) are at x = a and x = 3a respectively. Point charges +q and –q are located at x = 2a and x = 0 respectively on the axis of cylinder. The electric flux through the curved surface of cylinder is (nearly)

a) $0.6\frac{q}{\varepsilon_0}$ b) $1.8\frac{q}{\varepsilon_0}$ c) $0.9\frac{q}{\varepsilon_0}$ d) $\frac{q}{\varepsilon_0}$

7. Two metallic solid spheres of radii R and 2R are charged such that both of them have same charge density σ. If the spheres are located far away from each other and connected by a thin conducting wire, the new charge density on bigger sphere is :

a) 5σ b) 6σ c) $\frac{5}{6}\sigma$ d) 2σ

8. If radius of a hollow metallic sphere is 'R'. If the P.D between it's surface and a point at a distance 3R from it's centre is V then the electric field intensity at a distance 3R from it's centre is

a) $\frac{V}{2R}$ b) $\frac{V}{3R}$ c) $\frac{V}{4R}$ d) $\frac{V}{6R}$

9. A solid conducting sphere having a charge 'Q' is surrounded by an uncharged concentric conducting hollow spherical shell. Let the P.D between the surface of the solid sphere and that of the outer surface of the hollow shell be V. If the shell is now given a charge of –3Q. The new PD between the same two surfaces is

a) V b) 2V c) 4V d) –2V

10. Two concentric spherical shells of radii R and 2R carry charges Q and 2Q respectively. Change in electric potential on the outer shell when both are connected by a conducing wire is ($K = \frac{1}{4\pi\varepsilon_0}$)

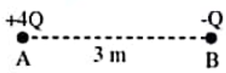
a) zero b) $\frac{3KQ}{2R}$ c) $\frac{KQ}{R}$ d) $\frac{2KQ}{R}$

More than One correct answer Type Questions

11. Five balls numbered 1 to 5 are suspended using separate threads. Pairs (1,2), (2,4) and (4,1) show electrostatic attraction while pairs (2,3) and (4,5) show repulsion. Therefore ball 1 must be

a) positively charged b) negatively charged
c) neutral d) made of metal

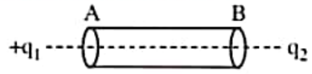
12. Two fixed charges 4Q (positive) and Q (negative) are located at A and B, the distance AB being 3 m.



a) The point P where the resultant field due to both is zero is on AB outside AB.
b) The point P where the resultant field due to both is zero is on AB inside AB.
c) If a positive charge is placed at P and displaced slightly along AB it will execute oscillations.
d) If a negative charge is placed at P and displaced slightly along AB it will execute oscillations.

3. Current Electricity

13. A thin conducting rod AB is introduced in between the two point charges $+q_1$ and $-q_2$ as shown in figure. For this situation mark the correct statement(s).



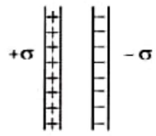
a) The total force experienced by q_1 is vector sum of electric force experienced by q_1 due to q_2 and due to induced charges on rod.

b) The end A will become negatively charged

c) The total force acting on $+q_1$, will be greater than as compared to the case without rod

d) The total force acting on $-q_2$ will be greater than as compared to the case without rod

14. Two infinite sheets of uniform charge density $+\sigma$ and $-\sigma$ are parallel to each other as shown in the figure. Electric field at the



a) points to the left or to the right of the sheets is zero.

b) midpoint between the sheets is zero.

c) midpoint of the sheets is σ/ε_0 and is directed towards right.

d) midpoint of the sheet is $2\sigma/\varepsilon_0$ and is directed towards right.

15. An electric field converges at the origin whose magnitude is given by the expressionE = 100r N/C, where r is the distance measured from the origin.

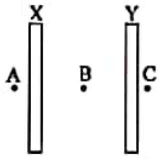
a) total charge contained in any spherical volume with its centre at origin is negative.

b) total charge contained at any spherical volume, irrespective of the location of its centre, is negative.

c) total charge contained in a spherical volume of radius 3 cm with its centre at origin has magnitude 3×10^{-13}C.

d) total charge contained in a spherical volume of radius 3 cm with its centre at origin has magnitude 3×10^{-9} Coul.

16. X and Y are large, parallel conducting plates closed to each other. Each face has an area A. X is given a charge Q. Y is without any charge. Points A, B and C are as shown in figure.



a) The field at B is $\frac{Q}{2\varepsilon_0 A}$

b) The field at B is $\frac{Q}{\varepsilon_0 A}$

c) The fields at A, B and C are of the same magnitude.

d) The field at A and C are of the same magnitude, but in opposite directions.

3. Current Electricity

17. Mark the correct options:

a) Gauss's law is valid only for uniform charge distributions.

b) Gauss's law is valid only for charges placed in vacuum.

c) The electric field calculated by Gauss's law is the field due to all the charges .

d) The flux of the electric field through a closed surface due to all the charges is equal to the flux due to the charges enclosed by the surface.

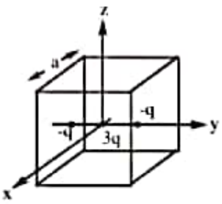
*18. A cubical region of side 'a' has its centre at the origin. It encloses three fixed point charges of charge -q at (0, -a/4,0), +3q at (0, 0, 0) and –q at (0, +a/4, 0). Choose the correct options(s)

a) The net electric flux crossing the plane x = +a/2 is equal to the net electric flux crossing the plane x = – a/2

b) The net electric flux crossing the plane y = +a/2 is more than the net electric flux crossing the plane y = – a/2

c) The net electric flux passing through the given cube is $\frac{q}{\varepsilon_0}$

d) The net electric flux crossing the plane z = +a/2 is equal to the net electric flux crossing the plane x = +a/2



19. Four charges of 1 mC, 2 mC, 3 mC, and – 6mC are placed one at each corner of the square of side 1m. The square lies in the x-y plane with its centre at the origin.

a) The electric potential is zero at the origin.

b) The electric potential is zero everywhere along the x-axis only of the sides of the square are parallel to x and y axis.

c) The electric potential is zero everywhere along the z-axis for any orientation of the square in the x-y plane.

d) The electric potential is not zero along the z-axis except at the origin.

20. Three point charges Q, 4Q and 16Q are placed on a straight line 9 cm long. Charges are placed in such a way that the system has minimum potential energy. Then

a) 4Q and 16Q must be at the ends and Q at a distance of 3 cm from the 16Q.

b) 4Q and 16Q must be at the ends and Q at a distance of 6 cm from the 16Q.

c) Electric field at the position of Q is zero.

d) Electric field at the position of Q is $\frac{Q}{4\pi\varepsilon_0}$.

21. Potential at a point A is 3 volt and at a point B is 7 volt, an electron is moving towards A from B.

a) It must have some K.E. at B to reach A

b) It need not have any K.E. at B to reach A

c) to reach A it must have more than or equal to 4 eV K. E. at B.

d) when it will reach A, it will have K.E. more than or at least equal to 4 eV if it was released from rest at B.

22. At distance of 5cm and 10cm outwards from the surface of a uniformly charged solid sphere, the potentials are 100V and 75V respectively. Then

a) potential at its surface is 150V.

b) the charge on the sphere is $(5/3) \times 10^{-10}$C.

c) the electric field on the surface is 1500 V/m.

d) the electric potential at its centre is 225V.

23. A point electric dipole with dipole moment 'P' is placed in the external uniform electric field whose strength is 'E_0' with 'p' parallel to 'E_0'. Then

a) Net force experienced by the dipole is zero

b) One of the equi-potential surfaces enclosing the dipole forms sphere

c) The radius of spherical equi-potential surface is $\left[\frac{P}{4\pi\varepsilon_0 E}\right]^{\frac{1}{3}}$

d) The radius of spherical equi-potential surface is $\sqrt{\frac{P}{4\pi\varepsilon_0 E}}$

24. An electric dipole moment $\vec{p}=(2.0\hat{i}+3.0\hat{j})\ \mu$C. m is placed in a uniform electric field $\vec{E}=(3.0\hat{i}+2.0\hat{k})\times 10^5$ N C^{-1}.

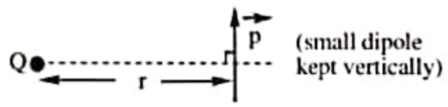
a) The torque that $\vec{E}$ exerts on $\vec{p}$ is $(0.6\hat{i}-0.4\hat{j}-0.9\hat{k})$ Nm.

b) The potential energy of the dipole is –0.6 J.

c) The potential energy of the dipole is 0.6 J.

d) If the dipole is rotated in the electric field, the maximum potential energy of the dipole is 1.3 J.

25. For the situation shown in the figure (assume r >> length of dipole), select the correct statement(s) [$\vec{p}$ = dipole moment, Q = charge on the particle which is on equatorial line of dipole)



a) Force acting on the dipole is zero

b) Force acting on the dipole is approximately $\frac{PQ}{4\pi\varepsilon_0 r^3}$ and is acting upwards

c) Torque acting on the dipole is $\frac{PQ}{4\pi\varepsilon_0 r^2}$ in clockwise direction

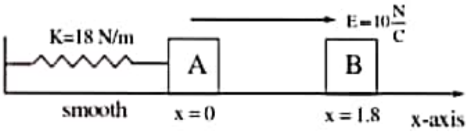
d) Torque acting on the dipole is $\frac{PQ}{4\pi\varepsilon_0 r^2}$ in anti-clockwise direction

Linked Comprehension Type Questions

Passage - I :

Electrostatic force on a charged parricle is given by $\vec{F}=q\vec{E}$. If q is positive $\vec{F}\uparrow \Rightarrow \vec{E}\uparrow$ and if q negative $\vec{F}\uparrow \Rightarrow \vec{E}\downarrow$

Question : In the figure $m_A = m_B = 1kg$. Block A is neutral while $q_B = -1C$. Sizes of A and B are negligible. B is released from rest at a distance 1.8 m from A. Initially spring is neither compressed nor elongated



26. If collision between A and B is perfectly inelastic, what is velocity of combined mass just after collision ?

a) 6 m/s b) 3 m/s c) 9 m/s d) 12 m/s

27. Equilibrium position of the combined mass is at x = ...m.

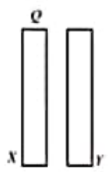
a) $-\frac{2}{9}$ b) $-\frac{1}{3}$ c) $-\frac{5}{9}$ d) $-\frac{7}{9}$

28. The amplitude of oscillation of the combined mass will be :

a) $\frac{2}{3}$m b) $\frac{\sqrt{124}}{3}$m c) $\frac{\sqrt{72}}{9}$m d) $\frac{\sqrt{106}}{9}$m

Passage - II :

Two conducting plates X and Y, each having large surface area A (on one side), are placed parallel to each other as shown in figure.



The plate X is given a charge Q whereas the other is neutral.

29. Find the surface charge density at the inner surface of the plate X.

a) $\frac{Q}{3A}$ b) $-\frac{Q}{A}$ c) $+\frac{Q}{A}$ d) $\frac{Q}{2A}$

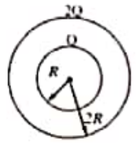
30. Find the electric field at a point to the left of the plates.

a) $\frac{Q}{2A\varepsilon_0}$ towards left b) $\frac{Q}{A\varepsilon_0}$ away from the plates

c) $\frac{QA}{\epsilon_0}$ towards left d) $\frac{Q}{\epsilon_0 A}$ towards the plates

Passage - III :

Statement : Inside a charged conducting spherical shell electric potential at any point is $\frac{Kq}{R}$ and at a point outside it, field is $\frac{Kq}{r^2}$ .Electric field inside the shell at any point is zero and outside it,field *is $\frac{Kq}{r^2}$. Here K is $\frac{1}{4\pi\varepsilon_0}$ R. is the radius of shell and r the distance from centre of shell. Two concentric spherical shells of radii R and 2R have charges Q and 2Q as shown in figure.*



31. If we draw a graph between potential V and distance r from the centre, the graph will be like

32. If we draw a graph between electric field E and distance r from the centre, it will be like :

33. Choose the correct option

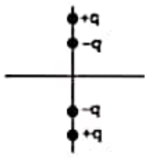
a) At a distance r (R < r < 2R) from the centre electric potential is $\frac{KQ}{R}$

b) At the same distance electric field is $\frac{KQ}{r^2}$

c) Both (a) and (b) are correct   d) Both (a) and (b) are wrong

Passage - IV :

Two insulated light rods of length l and 2l are placed in xy plane such their mid point is origin and they are free to rotate in xy plane about z-axis. Two + q charges are fixed at two ends of bigger rod and two - q charges are fixed at two ends of smaller rod.



34. What is electric dipolemoment of system ?

a) $3q\frac{l}{2}$ b) $5q\frac{l}{2}$ c) ql   d) Zero

35. Electric field at point (a, 0, 0) is E. Now if a >> l then

a) $E\alpha\frac{1}{a^2}$ b) $E\alpha\frac{1}{a^3}$ c) $E\alpha\frac{1}{a^4}$ d) $E\alpha\frac{1}{a^5}$

36. Work done to rotate smaller rod through 180^0 about z-axis

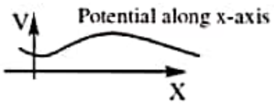
a) 0   b) $\frac{kq^2}{l}$ c) $\frac{2kq^2}{l}$ d) $\frac{3kq^2}{l}$

Matrix Matching Type Questions

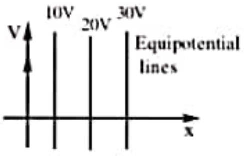
37. Electric field due to following configurations having charge density ρ

Column - I	Column - II
A) Infinite plane sheet of charge	p) 0
B) Infinite plane sheet of uniform thickness	q) $\frac{\rho}{2\varepsilon_0}$
C) Non-conducting charged solid sphere of radius R at its surfaces	r) $\frac{R\rho}{3\varepsilon_0}$
D) Non-conducting charge solid sphere of radius R at its center	s) $\frac{\rho}{\varepsilon_0}$

38. Column - I   Column - II

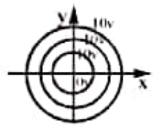
A) 

p) Field at origin is zero

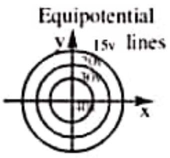
B) 

q) Zero field at some point

s) other than origin

C) 

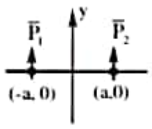
r) Field at +x immediately after origin is in the −ve x – direction

D) 

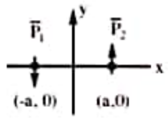
s) Field at +x immediately after origin is in the +ve x – direction.

39. In each situation of Column - I, two electric dipoles having dipole moments $\vec{p}_1$ and $\vec{p}_2$ of same magnitude (that is, $p_1 = p_2$) are placed on x-axis symmetrically about origin in different orientations as shown. In column - II certain inferences are drawn for these two dipoles. Then match the different orientations of dipoles in column - I with the corresponding results in column - II. [In column I the co-ordinates corresponding to the centres of dipoles and dipoles having the same dipole length]

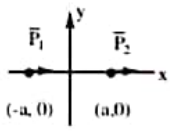
Column - I

A) 

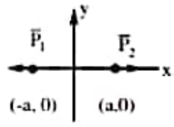
($\vec{p}_1$ and $\vec{p}_2$ perpendicular to x-axis as shown)

B) 

($\vec{p}_1$ and $\vec{p}_2$ and perpendicular x-axis as shown)

C) 

($\vec{p}_1$ and $\vec{p}_2$ and in the same x-y direction are parallel to x-axis as shown)

D) 

($\vec{p}_1$ and $\vec{p}_2$ and are opposite)

Column - II

p) The torque on one dipole due to other is zero.

q) The potential energy of one dipole in the electric field of other dipole is negative

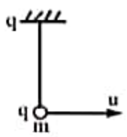
r) There is one straight line in plane (not at infinity) which is equipotential

s) Electric field at origin is zero.

Integer Type Questions

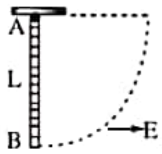
40. Three equal charges are placed at the three corners ABC of a square ABCD. If the force between the charges at A&B (on q_1 and q_2) is F_{12} and that between q_1 and q_3 is F_{13} the ratio of magnitudes F_{12}/F_{13} is ___

41. The bob of a pendulum has mass m = 1kg and charge $q = 40\mu C$. Length of pendulum is $l = 0.9$m. The point of suspension also has the same charge $40\mu C$. What is the minimum speed u (in m/s) should be imparted to the bob so that it can complete vertical circle?

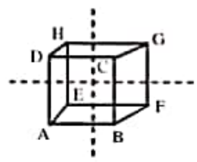


42. A 0.50 gm ball carries a charge of magnitude $10\mu C$. It is suspended by a string in a downward electric field of intensity 300 N/C. If the charge on the ball is positive, then the tension in the string is multiplies of 10^{-3} N is : ($g=10\ ms^{-2}$)

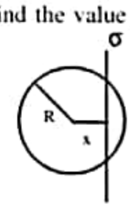
43. A uniform rod AB of length L and mass m is uniformly charged with a charge Q and it is freely suspended from end A as shown in figure. An eletric field E is suddenly switched on in the horizontal direction due to which rod get turned by a maximum angle 90°. The magnitude of E is how many number of times of $\frac{Mg}{Q}$



44. Two infinite line charges, each having a uniform charge density λ, pass through the midpoints of two pairs of opposite faces of a cube of edge L as shown in figure. The modulus of the total electric flux due to both the line charges through the face is ABCD is $\lambda L/(k\epsilon_0)$ find the value of k ?



45. An infinite, uniformly charged sheet with surface charge density σ cuts through a spherical Gaussian surface of radius R at a distance x from its center, as shown in the figure. The electric flux ϕ through the Gaussian surface is $\frac{k\pi(R^2-x^2)\sigma}{\varepsilon_0}$ find the value of k ?

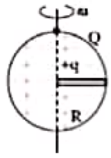


46. Find the potential difference V_{AB} between A (2m, 1m, 0m) and B (0m, 2m, 4m) in an electric field, $E=(x\hat{i}-2y\hat{j}+z\hat{k})V/m$ is ____ V

47. Uniform electric field of magnitude 100 V/m in space is directed along the line y = 3 + x. Find the potential difference between point A (3, 1) & B (1, 3)

48. A small ball of mass 1 kg and charge $\frac{2}{3}\mu C$ is placed at the centre of a uniformly charged sphere of radius 1 m and charg $\frac{1}{3}mC$. A narrow smooth horizontal groove is made in the sphere from centre

to surface as shown in figure. The sphere is made to rotate about its vertical diameter at a constant rate of $\frac{1}{2\pi}$ revolutions per second. Find the speed w.r.t. ground (in m/s) with which the ball slides out from the groove. Neglect any magnetic force acting on ball.



49. A dipole of dipole moment $\vec{P} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ is placed at point A (2, - 3, 1). The electric potential due to this dipole at the point B (4, – 1, 0) is equal to (All the parameters specified here are in S.I. units) ___$\times 10^9$ volts

50. A small electric dipole is kept on the axis of a uniformly charged ring at distance $R/\sqrt{2}$ from the centre of the ring. The direction of the dipole moment is along the axis. The dipole moment is P, charge of the ring is Q and radius of the ring is R. The force on the dipole is nearly ____.

KEY SHEET

1) b	2) d	3) a	4) a	5) c	6) a	7) c	8) d	9) a	10) a
11) cd	12) ad	13) a	14) ac	15) abc	16) acd	17) cd	18) acd	19) acd	20) bc
21) ac	22) abcd	23) abcd	24) abd	25) bc	26) b	27) c	28) d	29) d	30) a
31) c	32) a	33) b	34) d	35) a	36) a	37) A-q; B-q; C-r; D-p			
38) A-pqr; B-r; C-pq; D-ps			39) A-pr; B-pqrs; C-pqr; D-ps				40) 2		
41) 6	42) 8	43) 1	44) 2	45) 1	46) 3	47) 0	48) 2	49) 2	50) 0